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ORIGINAL RESEARCH

# Conditional differential analysis on the KATAN ciphers based on deep learning

Dongdong Lin[1,2] | Manman Li[1,2] | Zezhou Hou[1] | Shaozhen Chen[1,2]

[1]Information Engineering University, Zhengzhou, China

[2]State Key Laboratory of Cryptology, Beijing, China

**Correspondence**

Manman Li, Information Engineering University, Zhengzhou, China.
Email: limanman15@163.com

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**Abstract**

KATAN ciphers are block ciphers using non-linear feedback shift registers. In this study, the authors improve the results of conditional differential analysis on KATAN by using deep learning. Multi-differential neural distinguishers are built to improve the accuracy of the neural distinguishers and increase the number of its rounds. Moreover, a conditional differential analysis framework is proposed based on deep learning with the multi-differential neural distinguishers, resulting in a significant improvement than the previous. We present a practical key recovery attack on the 97-round KATAN32 with $2^{15.5}$ data complexity and $2^{20.5}$ time complexity. The attack of the 82-round KATAN48 and 70-round KATAN64 are also presented as the best known practical results.

**KEYWORDS**

block cipher, conditional differential analysis, deep learning, KATAN family, neural network

## 1 | INTRODUCTION

As an important part of Artificial intelligence, deep learning has been widely applied in image recognition, natural language processing, machine translation and other areas to solve a series of challenging problems [1]. Deep learning was proposed by Hinton in 2006 [2], which used neural networks to automatically extract features from the data and made feature engineering efficient and concise. The convolutional neural networks [3] and recurrent neural networks [4] had greatly enhanced the learning ability of neural networks.

With the widespread application of deep learning, researchers have proposed the assumption of combining deep learning with cryptanalysis. In 1991, Rivest [5] took cryptography and machine learning as a theoretically related area. In 2017, Greydanus [6] simulated the decryption function of the Enigma cipher using Long Short-Term Memory. At ICLR 2018, Aidan et al. [7] proposed a method of using generative adversarial networks to decrypt the Vigenère cipher. At CRYPTO 2019, Gohr [8] applied deep learning to the analysis of block ciphers for the first time to present a powerful distinguisher based on the neural network. The distinguisher improved the 11-round key recovery attack of Speck32/64. In 2021, Bao et al. [9] introduced the generalised neutral bits into Gohr's key recovery framework and first proposed a practical key recovery attack of 13-round Speck32/64 and 16-round Simon32/64. At EUROCRYPT 2021, Benamira et al. [10] presented a convincing explanation of neural network distinguishers, showing that the neural distinguishers rely on the differential distribution of ciphertext pairs and the differential distribution in the penultimate and antepenultimate rounds.

The KATAN family is a block cipher based on non-linear feedback shift registers (NLFSRs), designed by Cannière et al. [11] and published at CHES 2009. It is a hardware-oriented lightweight block cipher to provide sufficient security while realising the block size is small enough. In the single-key setting, Knellwolf et al. [12] first proposed a conditional differential (CD) analysis method of KATAN ciphers at ASIACRYPT 2010. Cube and algebraic attacks are also presented as practical attacks of KATAN ciphers [13]. Xing et al. [14] improved the conditional differential analysis of KATAN32. Additionally, there are some theoretical attack results of KATAN family [15–17]. We focus on improving conditional differential analysis by using deep learning. We call it 'conditional differential analysis based on deep learning (CDDL)'. In the single-key setting, our work is the best result in the practical key recovery attack on KATAN48 and KATAN64 as known so far. All the results are summarised in Table 1.

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## 1.1 | Our Contributions

In this study, we present a conditional differential analysis method based on deep learning. First, we analyse the basic principle of neural distinguishers and propose multi-differential neural distinguishers, which improve the accuracy and number of rounds of the distinguishers. Based on the new data format, a neural network structure is designed. Second, conditional neural distinguishers are proposed, combined with the multi-differential neural distinguishers and conditions. The message modification technology reduces the computational complexity at the plaintext filtering stage and the key search strategy reduces the computational complexity at the key recovery stage. Third, a conditional differential analysis framework is proposed. We present practical key recovery attacks against 97/82/70 rounds KATAN32/48/64, which can recover 30 continuous subkey bits. Compared to Ref. [12], our work not only increases the number of attack rounds and the recoverable key bits but also reduces the computational complexity.

**TABLE 1** Summary of single-key attacks on KATAN ciphers

| Cipher | Technique | Rounds | Time | Data | Reference |
|---|---|---|---|---|---|
| KATAN32 | CD | 78 | $2^{22}$ | $2^{22}$ CP | [12] |
| | Cube | 60 | $2^{39}$ | $2^{30.3}$ CP | [13] |
| | Algebraic | 79 | 14.7 min | 20 CP | [13] |
| | CDDL | 97 | $2^{20.5}$ | $2^{15.5}$ CP | Section 5.3 |
| | Improved CD | 97 | $2^{30}$ | $2^{19}$ CP | [14] |
| | | 98 | $2^{31}$ | $2^{19}$ CP | [14] |
| | | 99 | $2^{33}$ | $2^{19}$ CP | [14] |
| | Differential(*) | 95 | $2^{78}$ | - | [15] |
| | | 112 | $2^{78}$ | - | [15] |
| | | 114 | $2^{78}$ | - | [15] |
| | | 115 | $2^{78}$ | - | [15] |
| | Dynamic cube(*) | 118 | $2^{78.3}$ | $2^{19}$ CP | [16] |
| | | 138 | $2^{78.5}$ | $2^{31}$ CP | [16] |
| | | 155 | $2^{78.3}$ | $2^{32}$ CP | [16] |
| | MITM(*) | 201 | $2^{78.1}$ | 3 KP | [17] |
| | | 206 | $2^{79}$ | 3 KP | [17] |
| KATAN48 | CD | 70 | $2^{34}$ | $2^{34}$ CP | [12] |
| | Cube | 40 | $2^{49}$ | $2^{25}$ CP | [13] |
| | Algebraic | 64 | 6.4 h | 5 CP | [13] |
| | CDDL | 82 | $2^{18.9}$ | $2^{14.2}$ CP | Section 6.1 |
| | MITM(*) | 146 | $2^{78.1}$ | 2 KP | [17] |
| | | 148 | $2^{79}$ | 2 KP | [17] |
| KATAN64 | CD | 68 | $2^{35}$ | $2^{35}$ CP | [12] |
| | Cube | 30 | $2^{35}$ | $2^{20.6}$ CP | [13] |
| | Algebraic | 60 | 3.2 h | 5 KP | [13] |
| | CDDL | 70 | $2^{20.2}$ | $2^{16.7}$CP | Section 6.2 |
| | MITM(*) | 126 | $2^{78.1}$ | 2 KP | [17] |
| | | 129 | $2^{79}$ | 2 KP | [17] |

*Note*: *theoretical attacks.

Abbreviations: CP, chosen plaintext; KP, known plaintext; MITM, meet-in-the-middle.

## 1.2 | Outline

The remainder of the structure is as follows: Section 2 provides some preliminaries, a brief description including the specification of KATAN ciphers and differential analysis based on deep learning. Section 3 introduces the construction of multi-differential neural distinguishers in detail. Section 4 proposes a framework of conditional differential analysis based on deep learning. We present practical key recovery attacks against KATAN ciphers in Section 5 and Section 6. The summary is in Section 7.

# 2 | PRELIMINARIES

## 2.1 | Notation

| | |
|---|---|
| $P$, $C$: | plaintext, ciphertext; |
| $P^{block}$, $C^{block}$: | a $P^{block}$ ($C^{block}$) consists of 32 plaintext (ciphertext) pairs; |
| $P_{structure}$, $C_{structure}$: | a $P_{structure}$ ($C_{structure}$) consists of $2^s$ $P^{block}$ ($C^{block}$); |
| $L_t$, $R_t$: | the states of two NLFSRs at round $t$; |
| $\Delta P$, $\Delta C$, $\Delta L_t$, $\Delta R_t$: | the differences of plaintext, ciphertext, $L_t$ and $R_t$; |
| $p_i$, $c_i$, $l_{t+i}$, $r_{t+i}$: | the $i$th bit of plaintext, ciphertext, $L_t$ and $R_t$; |
| $K$, $k_i$: | the master key, the $i$th bit of the key. |

## 2.2 | Description of KATAN

KATAN ciphers have three versions: KATAN32, KATAN48 and KATAN64. The three versions have block sizes of 32, 48 and 64 bits. The family of ciphers shares the same key schedule, master key length, encryption rounds and NLFSRs.

The master key $K = (k_0, \ldots, k_{79})$ of KATAN family is 80 bits. The key schedule uses an 80-bit linear feedback shift register (LFSR) and the feedback polynomial is an 80-bit primitive polynomial.

$$k_{i+80} = k_i \oplus k_{i+19} \oplus k_{i+30} \oplus k_{i+67}, \quad 0 \le i \le 427$$

The structure of KATAN is shown in Figure 1.

The round functions in each round are two non-linear functions $f_1$ and $f_2$ which are defined as follows:

$$f_1[R] = R[x_1] \oplus R[x_2] \oplus (R[x_3]R[x_4]) \oplus (R[x_5]R[x_6]) \oplus k_{2t+1}$$

$$f_2[L] = L[y_1] \oplus L[y_2] \oplus (L[y_3]L[y_4]) \oplus L[y_5]a_t \oplus k_{2t}$$

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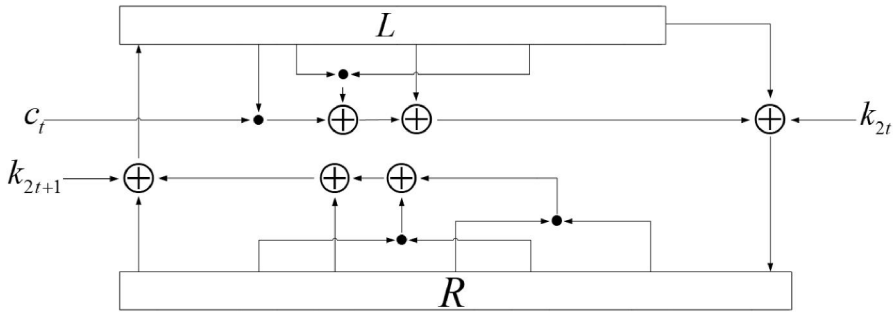
**FIGURE 1** The structure of KATAN




where $a_t$ is a round constant which has the initial value (1, 1, 1, 1, 1, 1, 1, 0) and is updated by a feedback polynomial of $a_{t+8} = a_t + a_{t+1} + a_{t+3} + a_{t+5}$, $t \geq 8$. $k_{2t+1}$ and $k_{2t}$ are the two key bits. The definitions of indices $\{x_i\}$ and $\{y_i\}$ are different in various versions. KATAN48 applies the functions $f_1$ and $f_2$ twice by using the same key bits in one round. For KATAN64, $f_1$ and $f_2$ are applied three times each round. The differences between KATAN ciphers are shown in Table 2. After 254 rounds of encryption, the states in the registers $L$ and $R$ are outputted as the ciphertext.

Taking KATAN32 as an example, the block size of KATAN32 is 32 bits. The lengths of registers $L$ and $R$ are 13 and 19 bits. The plaintext is loaded into $L$ and $R$:

$$(p_{31}, p_{30}, \ldots, p_{19}) \rightarrow (l_0, l_1, \ldots, l_{12})$$

$$(p_{18}, p_{17}, \ldots, p_0) \rightarrow (r_0, r_1, \ldots, r_{18})$$

For KATAN32, $f_1$ and $f_2$ are updated according to the following two equations at round $t$:

$$l_{t+13} = r_t \oplus r_{t+11} \oplus (r_{t+6} r_{t+8}) \oplus (r_{t+10} r_{t+15}) \oplus k_{2t+1}$$

$$r_{t+19} = l_t \oplus l_{t+5} \oplus (l_{t+4} l_{t+7}) \oplus l_{t+9} a_t \oplus k_{2t}$$

where $0 \leq t \leq 253$, $l_{t+13}$ and $r_{t+19}$ are the updated bits of $L_t$ and $R_t$.

**TABLE 2** KATAN family parameters

| Cipher | $\|L\|$ | $\|R\|$ | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $x_6$ |
|---|---|---|---|---|---|---|---|---|
| KATAN32 | 13 | 19 | 0 | 11 | 6 | 8 | 10 | 15 |
| KATAN48 | 19 | 29 | 0 | 9 | 7 | 15 | 13 | 22 |
| KATAN64 | 25 | 39 | 0 | 13 | 15 | 17 | 24 | 29 |
| **Cipher** | $y_1$ | $y_2$ | $y_3$ | $y_5$ | $y_5$ | | | |
| KATAN32 | 0 | 5 | 4 | 7 | 9 | | | |
| KATAN48 | 0 | 6 | 3 | 11 | 12 | | | |
| KATAN64 | 0 | 9 | 4 | 13 | 15 | | | |

## 2.3 | Description of differential analysis based on deep learning

At CRYPTO 2019, Gohr proposed a differential analysis framework based on deep learning for the first time. The neural distinguishers of the round-reduced Speck32/64 are constructed based on the neural network and trained with the input difference and ciphertext pairs. At the key recovery stage, combined with the Bayesian optimization algorithm [18], Gohr proposed an 11-round Speck32/64 key recovery attack by using neural distinguishers.

First, we introduce Gohr's construction method of neural distinguishers. The input difference is $\Delta = 0x00400000$. The plaintext pair and ciphertext pair are $(P, P')$ and $(C, C')$. The input of the neural network is ciphertext pairs. Generate 0–1 data as label $Y$ for ciphertext pairs randomly. The value of $P'$ is decided by label Y. If $Y = 1$ and $P' = P \oplus \Delta$. If $Y = 0$, $P'$ is generated randomly. Put the ciphertext pairs and their labels into the neural distinguisher. If the distinguisher can correctly distinguish the labels with an accuracy of $\geq$50%, a useable neural distinguisher is trained. Gohr trained neural distinguishers of the round-reduced Speck32/64 by using a deep residual neural network [19]. The accuracy of neural distinguishers decreases while the number of rounds increases. Obviously, the higher the accuracy of the neural distinguishers is, the stronger the distinguishing ability is. The computational complexity of the key recovery stage will also decrease with the enhancement of the distinguishing ability. Therefore, it is meaningful to improve the accuracy of the neural distinguishers by modifying the network structure and data format.

Gohr performed a key recovery attack on the $r + 1$-round Speck32/64 by using an $r$-round neural distinguisher. Use the random key $K_r$ to decrypt the ciphertext of the $r + 1$-round Speck32/64 for one round. Score and rank the current random key $K_r$ with the $r$-round neural distinguisher. The score function is as follows:

$$v_{K_r} = \sum_{i=0}^{n-1} \log_2 \left( \frac{Z_i^{K_r}}{1 - Z_i^{K_r}} \right)$$

where $n$ is the number of ciphertext pairs. $Z_i^{K_r}$ is the response value of the neural distinguisher to the ciphertext pair decrypted with the guess key $K_r$. When the score of the guess key $K_r$ exceeds the threshold $c_1$, the current guess key $K_r$ is returned.

The subkeys of Speck32/64 are 16 bits in one round. The computation cost of scoring $2^{16}$ keys is too high. To improve the efficiency of the key recovery attack, Gohr used Bayesian

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optimisation algorithm to optimise the key search policy. The non-random structure of the wrong key response value distribution generated the wrong key response profile. New candidate keys are generated in iterations of the key search. To perform a key recovery attack with more rounds, Gohr had to use neural distinguishers with an accuracy of almost 50%, which made the response of the distinguishers weak. To enhance the distinguishers' response, Gohr used $s$ neutral bits [20] to generate a plaintext structure consisting of $2^s$ plaintext pairs, which passed the first two rounds differential together. The plaintext structure is encrypted to obtain the resulting ciphertext structure. Gohr partially decrypted each ciphertext pair by using all final-round subkeys and the response value of ciphertext structure to rank all subkeys. Using the above key recovery attack strategy, Gohr presented an 11-round Speck32/64 key recovery attack.

## 3 | MULTI-DIFFERENTIAL NEURAL DISTINGUISHERS

The core idea of conditional differential analysis based on deep learning is to train neural distinguishers. A well-behaved neural distinguisher can effectively increase the rounds of the key recovery attack. At the same round, the higher the accuracy of the neural distinguisher is, the lower the computational complexity of the key recovery attack is. In Ref. [10], Benamira et al. demonstrated that the neural distinguishers relied on the differential distributions, which inspired us to construct multi-differential neural distinguishers. Using multi-differential data, we improve the accuracy of the neural distinguishers and increase its number of rounds. Based on the multi-differential data format, we design a neural network structure based on the KATAN ciphers.

### 3.1 | Neural network structure

Gohr used a deep residual neural network to obtain a round-reduced Speck32/64 neural distinguisher. Experiments verified that the deep residual neural network is universal to a certain extent for different algorithms. In this section, we design a deep residual neural network based on the data format of KATAN ciphers. The specific structure is shown in Figure 2.

The neural network consists of three parts: the input layers, the iterative layers and the output layers. The input layers receive KATAN's ciphertext data and reshape the data into a matrix form. The initial convolution performs simple feature extraction on the data. In the iterative layers, the neural network uses five residual blocks. Each residual block consists of two Conv1D layers, which is followed by a batch normalisation layer and an activation layer. The output of the residual block and the input are added. The result is passed as the input for the next block. The output layer consists of a fully connected layer and an output unit, sigmoid.

The hyperparameters of the neural network are given below. The convolutional kernel is one in the input layers and three in the iterative layers. The number of filters in the convolutional layers is $n$ (block size). The rest of the hyperparameters are shown in Table 3.

### 3.2 | Data format of multi-differential neural distinguishers

Since deep learning is a data-driven technology, the quality of data plays a key role in the quality of the neural model. For neural distinguishers, data format directly affects the accuracy. The deep residual network is originally used in the image recognition field, which has the advantage of recognising matrix data. We replace the ciphertext pairs with output differences and propose a multi-differential neural distinguisher. To take advantage of the characteristics of the deep residual network, we concatenate the output differences into a matrix. The neural network extracts not only the features of each output difference but also the connection between different output differences. The construction of multi-differential data is shown in Figure 3.

As shown in Figure 3, we generate $n$ plaintext pair sets $(P_{set}, P'_{set})$ randomly, where $P_{set} = (P_0, P_1, \ldots, P_{n-1})$ and

**TABLE 3** Neural network hyperparameters

| Hyperparameter | Value |
|---|---|
| Training size | $10^7$ |
| Validation size | $10^6$ |
| Batch size | 1000 |
| Epochs | 200 |
| Optimiser | Adam |
| Learning rate | $10^{-4}$ |
| Loss function | MSE (mean-squared-error) |

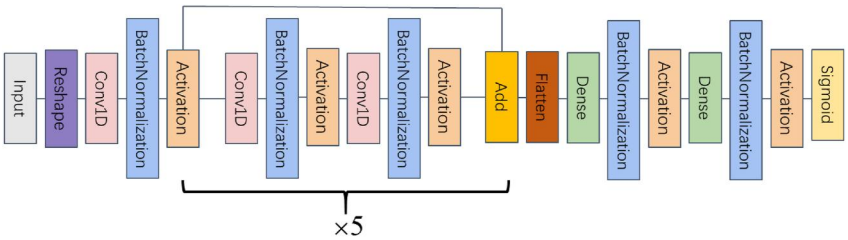


**FIGURE 2** Neural network structure

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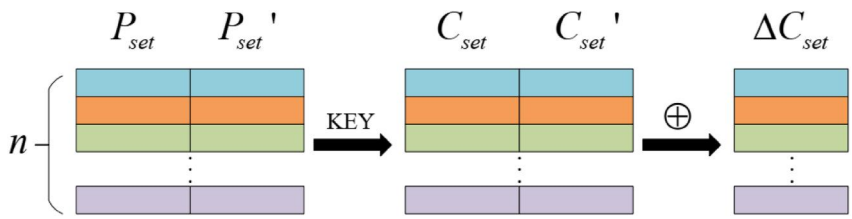


**FIGURE 3** Data format of multi-differential neural distinguishers

$P'_{set} = (P'_0, P'_1, \ldots, P'_{n-1})$. The input difference is $\Delta$. The resulting ciphertext pair sets and ciphertext differences are $(C_{set}, C'_{set})$ and $\Delta C_{\text{set}}$. The ciphertext differences are the input of the neural network. We generate 0–1 data as label $Y$ for ciphertext pair sets randomly, If $Y = 1$, $P'_i = P_i \oplus \Delta$, $i \in [0, n-1]$. If $Y = 0$, $P'_{set}$ is generated randomly.

In the following two experiments, we verify the advantages of multi-differential neural distinguishers. The experiments in this section are all performed on the round-reduced KATAN32. The input size of the neural network is $(n \times n, 1)$ and the hyperparameter settings are shown in Table 3. We compare the accuracy of neural distinguishers under different data format in our experiments, which are shown in Table 4. When the multi-difference is '-', which means that the data is ciphertext pair, the accuracy of the 51-round neural distinguisher is 53.3%. When the multi-difference $n = 1$, the accuracy is slightly higher than the neural distinguisher using ciphertext pairs. We find that the accuracy of multi-differential neural distinguishers is positively correlated with the number of multi-differences. When $n = 32$, the accuracy is improved by 21.4%. For the 58-round neural distinguisher, if the data is ciphertext pairs, we fail to train a distinguisher with an accuracy of ≥50%. When $n = 32$, we successfully train a distinguisher with an accuracy of 55.3%.

We find that the multi-differential neural distinguishers utilise more output differences in the matrix, allowing the neural network to learn more features. Comparing the same training and validation set size, the multi-differential neural distinguishers require $n$ times data than the neural distinguishers using ciphertext pair. The following experiments prove that the improvement of model accuracy has no connection with the increase of the data. We train a multi-differential neural distinguisher with $n = 32$ and a neural distinguisher using ciphertext pairs. The sizes of two train sets are $32 \times 10^6$ and $10^6$ while the sizes of two validation sets are $32 \times 10^5$ and $10^5$. The results are shown in Figure 4.

As shown in Figure 4, the improvement of model accuracy has no connection with the enhancement of the data. Based on the above experimental results, we use the multi-differential neural distinguisher with $n = 32$ to perform the key recovery attack of KATAN32.

## 4 | CONDITIONAL DIFFERENTIAL ANALYSIS BASED ON DEEP LEARNING

Conditional differential analysis prevents the propagation of differences by controlling the state of NLFSRs with conditions. The basic idea is as follows: Firstly, it searches the input difference and conditions, which can control the propagation of the differences. Secondly, the conditions are imposed to filter plaintexts which can lead to bias in some update bit differences. Thirdly, key recovery attacks are proposed using the bias.

**TABLE 4** the accuracy of neural distinguishers under different data format

| Cipher | Round | The number of multi-differences $n$ | Accuracy |
|---|---|---|---|
| KATAN32 | 51 | – | 53.3% |
| | 51 | 1 | 53.8% |
| | 51 | 4 | 58.6% |
| | 51 | 16 | 67.0% |
| | 51 | 32 | 74.7% |
| | 58 | – | 50.0% |
| | 58 | 32 | 55.3% |

Although conditional differential analysis increases the rounds of differential attack, only a few discontinuous key bits can be recovered. Based on the multi-differential neural distinguishers proposed in Section 3, we present conditional differential analysis based on deep learning.

### 4.1 | Conditions

For KATAN ciphers, the difference of updated bits in each round is determined by the partial bits and the differences of the previous round. Applying the conditions, the partial differential bits can be controlled to prevent the propagation of differences. Conditions are generally used to prevent differential propagation in the initial rounds. In Ref. [12], the conditions are classified into three types:

I. Conditions only involve plaintext bits $p_i$.
II. Conditions involve plaintext bits $p_i$ and key bits $k_i$.
III. Conditions only involve key bits $k_i$.

This paper uses the Mixed-Integer Linear Programming (MILP) model [14] to search the conditions of the KATAN ciphers, which also can be found in a manual operation. The input difference has an impact on the accuracy of the neural distinguishers. The multi-differential neural distinguishers require that the data satisfies the same input difference. When searching for the conditions of the initial rounds before neural

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distinguishers, a constraint to the output difference should be added. We find a suitable differential characteristic with probability 1.

## 4.2 | Key recovery attack

At the key recovery stage, we use the Bayesian key search strategy to reduce the computational complexity. The response value of the neural distinguishers for the wrong key depends on its bitwise difference to the correct key. The larger the bitwise differences are, the smaller the response values of the neural distinguishers are. Therefore, the Bayesian key search strategy can quickly search for good keys. We randomly generate a set of candidate keys. After partial decryption, we use the neural distinguishers to score the candidate keys. According to the scores and wrong key response profiles, a new set of candidate keys is generated. We loop the above steps for a few cycles to maximise the probability density of the response of the neural distinguishers. The wrong key response profiles are generated at the precomputation stage.

The wrong key response profiles can compare the responses of different wrong keys to the neural distinguishers, which are also the basis of the Bayesian key search strategy. We randomly generate a master key $K$ and its corresponding 16-bit subkeys $k^{16}$ in $t$ rounds. The input difference is $\Delta$. We generate $n$ (block size) plaintext pairs $(P, P')$ and encrypt $r + t$ rounds to generate $n$ ciphertext pairs $(C, C')$. For each $\delta \in GF(2)^{16}$, we calculate $k^{16\prime} = k^{16} \oplus \delta$ and decrypt $n$ ciphertext pairs $(C, C')$ using $k^{16\prime}$. With the $r$-round ciphertext difference, we get the response values of the $r$-round neural distinguisher. We generate 3000 keys and repeat the above steps to calculate the empirical mean and standard deviation of the response values. We call it '$r + t$-round wrong key response profile'.

Taking KATAN32 as an example, we describe the key recovery attacks. Using the method in Section 4.1, we find a differential characteristic for $l$ rounds and train a multi-differential neural distinguisher for $r$ rounds. We get an $R = (l + r)$-round conditional neural distinguisher.

As shown in Figure 5, we use the distinguisher to perform an $R + t$-round key recovery attack, which contains 16-bit keys. The input difference of plaintext is $\Delta P$. The 32 plaintext pairs are $(P, P')$, where $P' = P \oplus \Delta P$. A plaintext block $P^{\text{block}}$ consists of 32 $(P, P')$. We use $s$ neutral bits to create a plaintext structure $P_{\text{structure}}$ consisting of $2^s$ plaintext blocks. After $R + t$-round encryption, we obtain the resulting ciphertext structure $C_{\text{structure}}$. The Bayesian key search strategy first randomly generates a set of candidate keys and uses each candidate key to partially decrypt the ciphertext structure $C_{\text{structure}}$ to obtain the ciphertext differential structure $\Delta C_{\text{structure}}$. Then, we calculate the average of the conditional neural distinguisher responses to these candidate keys. We

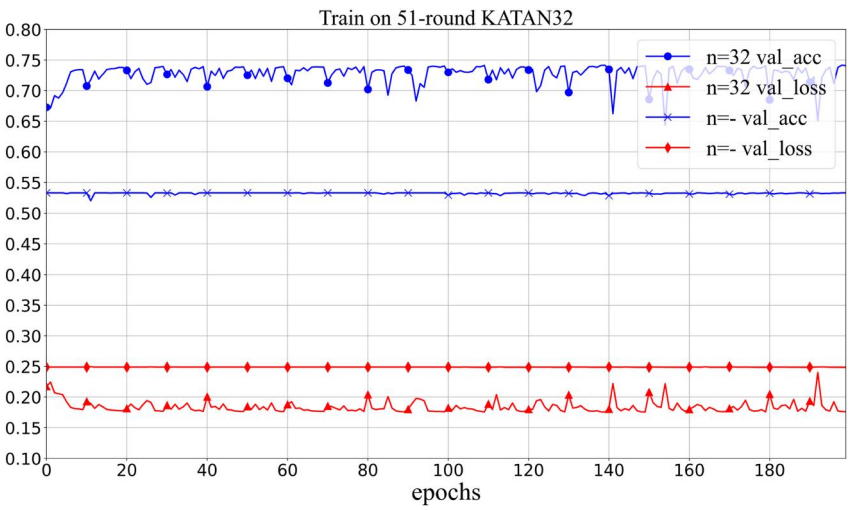


**FIGURE 4** The accuracy of multi-difference and ciphertext data for the same size of data set

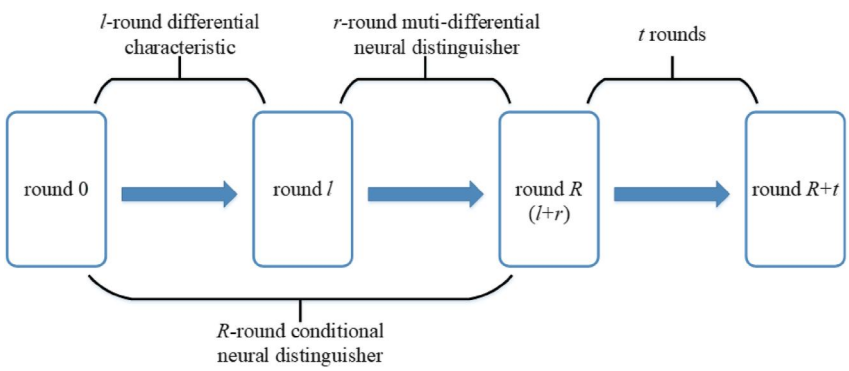


**FIGURE 5** $R + t$-round key recovery attack

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generate a new set of candidate keys using the scores and wrong key response profiles. We loop the above steps for the iteration numbers $z$. Algorithm 1 sums up the algorithm.

**Algorithm 1** $R + t$**-round Bayesian key search strategy**

**Require:** Ciphertext structure $C_{structure} = \{C_0^{block}, C_1^{block}, \ldots, C_{2^s-1}^{block}\}$, $R$-round conditional neural distinguisher $ND_R$, number of iterations $z$ and number of the set of candidate keys $n$. $M$ makes differences to the multi-differential data format.
$S := \{k_0^{16}, k_1^{16}, \ldots, k_{n-1}^{16}\} \leftarrow$ randomly choose the set of candidate keys without replacement.
$L \leftarrow \{\}$
**for** $j \in \{0, 1, \ldots, z-1\}$ **do**
  **for** $i \in \{0, 1, \ldots, 2^s - 1\}$ **do**
    **for** $u \in \{0, 1, \ldots, m-1\}$ **do**
      $P_{u,i,k^{16}}^{block} \leftarrow Decrypt_t_rounds\left(C_{u,i}^{block}, k^{16}\right)$, for all $k^{16} \in S$, $C_{u,i}^{block} \in C_i^{block}$
      $P_{u,i,k^{16}}^{block\prime} \leftarrow Decrypt_t_rounds\left(C_{u,i}^{block\prime}, k^{16}\right)$, for all $k^{16} \in S$, $C_{u,i}^{block\prime} \in C_i^{block}$
      $\Delta P_{u,i,k^{16}}^{block} = P_{u,i,k^{16}} \oplus P_{u,i,k^{16}}{}'$, for all $k^{16} \in S$
      $\Delta P_{i,k^{16}}^{block} \leftarrow M\left(\Delta P_{u,i,k^{16}}^{block}\right)$, for all $k^{16} \in S$
    $v_{i,k^{16}} \leftarrow ND_R\left(\Delta P_{i,k^{16}}^{block}, k^{16}\right)$, for all $k^{16} \in S$
    $w_{i,k^{16}} \leftarrow \log_2\left(v_{i,k^{16}}/\left(1 - v_{i,k^{16}}\right)\right)$, for all $k^{16} \in S$
  $w_{k^{16}} \leftarrow \sum_{i=0}^{2^s-1} w_{i,k^{16}}$, $k^{16} \in S$
  $L \leftarrow L\ c\ ups; [(k^{16}, w_{k^{16}}), k^{16} \in S]$
  $m_{k^{16}} \leftarrow \sum_{i=0}^{2^s-1} w_{i,k^{16}}/2^s$, $k^{16} \in S$
  $\lambda_{k^{16}} \leftarrow \sum_{i=0}^{n-1}\left(m_{k_i^{16}} - \mu_{k^{16} \oplus k_i^{16}}\right)^2 / \sigma^2_{k^{16} \oplus k_i^{16}}$, $k^{16} \in \{0, 1, \ldots, 2^{16} - 1\}$
  $S \leftarrow argsort_{k^{16}}(\lambda)[0 : n-1]$
**end for**
**return** $L$

If the response values of the candidate keys are higher than a cutoff threshold $c_1$, we present another $t$-round key recovery attack. For $R + t$-round key recovery attack, 14-bit continuous subkeys and 2-bit discontinuous subkeys are obtained. We use the 14-bit continuous subkeys to decrypt seven rounds and use an $R - 7$-round conditional neural distinguisher to repeat the above process. If the response values are higher than a cutoff threshold $c_2$, the present key guess is returned.

## 5 | KEY RECOVERY ATTACK OF KATAN32

In Section 4, using the multi-differential neural distinguishers, we solve the problem that conditional differential analysis can only recover a few discontinuous key bits. Using the Bayesian key search strategy, we reduce the computational complexity of the key recovery attack. Then, we apply the CDDL to present a practical key recovery attack of the 97-round KATAN32 in this section. For KATAN32, we need to choose an appropriate input difference to get a multi-differential neural distinguisher with more rounds. Based on the experimental experience, when the input difference is $\Delta = 0x00002000$, the multi-differential neural distinguishers have the highest accuracy under the same conditions. We use the MILP model to search for the smallest number of conditions and input difference which guarantees the

**TABLE 5** Differential characteristic and conditions of KATAN32

| Round | Difference state | Condition |
|---|---|---|
| 0 | 0110000100001 0000010001000000000 | |
| 1 | 1100001000010 0000100010000000000 | $p_{27} = 0$ |
| 2 | 1000010000100 0001000100000000001 | $p_{11} = 0$ |
| 3 | 0000100001000 0010001000000000010 | |
| 4 | 0001000010000 0100010000000000100 | $p_7 = 0, p_{21} = 1$ |
| 5 | 0010000100000 1000100000000001000 | |
| 6 | 0100001000000 0001000000000010000 | $p_3 = 1, p_{22} = 0$ |
| 7 | 1000010000000 0010000000000100000 | |
| 8 | 0000100000000 0100000000001000000 | |
| 9 | 0001000000000 1000000000010000000 | $l_{15} = 0$ |
| 10 | 0010000000000 0000000000100000000 | |
| 11 | 0100000000000 0000000001000000000 | $r_{25} = 0$ |
| 12 | 1000000000000 0000000010000000000 | |
| 13 | 0000000000000 0000000100000000001 | $p_0 = 0$ |
| 14 | 0000000000000 0000001000000000010 | |
| 15 | 0000000000000 0000010000000000100 | $r_{22} = 0$ |
| 16 | 0000000000000 0000100000000001000 | |
| 17 | 0000000000000 0001000000000010000 | $r_{26} = 0$ |
| 18 | 0000000000000 0010000000000100000 | |
| 19 | 0000000000000 0100000000001000000 | |
| 20 | 0000000000000 1000000000010000000 | |
| 21 | 0000000000000 0000000000100000000 | |
| 22 | 0000000000000 0000000001000000000 | $r_{36} = 0$ |
| 23 | 0000000000000 0000000010000000000 | |
| 24 | 0000000000000 0000000100000000000 | $r_{29} = 0$ |
| 25 | 0000000000000 0000001000000000000 | |
| 26 | 0000000000000 0000010000000000000 | $r_{33} = 0$ |

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output difference is $\Delta = 0x00002000$. A 26-round differential characteristic with probability 1 is found. We train the multi-differential neural distinguishers of KATAN32 with the input difference $\Delta$. Under these preparations, we present a key recovery attack of the 97-round KATAN32. The time complexity and data complexity of the key recovery attack are $2^{20.5}$ and $2^{15.5}$, respectively.

## 5.1 | Conditions of KATAN32

We use the MILP model to search for differential characteristics with probability 1 for KATAN32. A 26-round differential characteristic with 14 conditions has been searched. The differential characteristic and conditions are presented in Table 5.

The conditions in Table 5 are expressed by plaintext bits and key bits as follows:

$$l_{15} = p_{16} \oplus p_5 \oplus p_{10}p_8 \oplus p_6p_1 \oplus k_5 = 0,$$

$$r_{25} = p_{25} \oplus p_{20} \oplus p_{21}(p_{18} \oplus p_7 \oplus p_{12}p_{10} \oplus p_8p_3 \oplus k_1) \oplus (p_{16} \oplus p_5 \oplus p_{10}p_8 \oplus p_6p_1 \oplus k_5) \oplus k_{12} = 0,$$

$$r_{22} = p_{28} \oplus p_{23} \oplus p_{24}p_{21} \oplus p_{19} \oplus k_6 = 0,$$

$$r_{26} = p_{24} \oplus p_{19} \oplus p_{20}(p_{17} \oplus p_6 \oplus p_{11}p_9 \oplus p_7p_2 \oplus k_3) \oplus k_{14} = 0,$$

After imposing 14 conditions, a 26-round differential characteristic with $\Delta P = 0x61082200 \xrightarrow{prob.=1} \Delta = 0x00002000$ is found. We reshape the conditions to reduce the computational complexity of filtering the plaintexts and modify each randomly generated plaintext with the reshaped conditions. For example, condition $l_{15} = 0$ is reshaped as follows:

$$p_{16} = p_5 \oplus p_{10}p_8 \oplus p_6p_1 \oplus k_5$$

The final plaintexts satisfy all the conditions. The plaintext bits in the reshaped conditions can not be modified under the following conditions. To make all conditions modifiable, we add four conditions:

$$p_{20} = 1, p_6 = 0, l_{14} = p_{17} \oplus p_6 \oplus p_{11}p_9 \oplus p_7p_2 \oplus k_3 = 0,$$

$$l_{21} = p_{10} \oplus (p_{31} \oplus p_{26} \oplus p_{27}p_{24} \oplus p_{22} \oplus k_0) \oplus p_4p_2 \oplus p_0(p_{27} \oplus p_{22} \oplus p_{23}p_{20} \oplus (p_{18} \oplus p_7 \oplus p_{12}p_{10} \oplus p_8p_3 \oplus k_1) \oplus k_8) \oplus k_{15} = 1.$$

After imposing 18 conditions, we have a 26-round differential characteristic with $\Delta P = 0x61082200 \xrightarrow{prob.=1} \Delta = 0x00002000$.

To attack more rounds, we have to use multi-differential neural distinguishers with an accuracy of almost 50%, which respond weakly to wrong keys. To boost the response of

$$r_{36} = (p_{14} \oplus p_3 \oplus p_8p_6 \oplus p_4(p_{31} \oplus p_{26} \oplus p_{27}p_{24} \oplus p_{22} \oplus k_0) \oplus k_9) \oplus (p_9 \oplus (p_{30} \oplus p_{25} \oplus p_{26}p_{23} \oplus p_{21} \oplus k_2) \oplus p_3p_1 \oplus (p_{31} \oplus p_{26} \oplus p_{27}p_{24} \oplus p_{22} \oplus k_0)(p_{26} \oplus p_{21} \oplus p_{22}p_{19} \oplus (p_{17} \oplus p_6 \oplus p_{11}p_9 \oplus p_7p_2 \oplus k_3) \oplus k_{10}) \oplus k_{19}) \oplus (p_{10} \oplus (p_{31} \oplus p_{26} \oplus p_{27}p_{24} \oplus p_{22} \oplus k_0) \oplus p_4p_2 \oplus p_0(p_{27} \oplus p_{22} \oplus p_{23}p_{20} \oplus (p_{18} \oplus p_7 \oplus p_{12}p_{10} \oplus p_8p_3 \oplus k_1) \oplus k_8) \oplus k_{17})(p_7 \oplus (p_{28} \oplus p_{23} \oplus p_{24}p_{21} \oplus p_{19} \oplus k_6) \oplus p_1(p_{31} \oplus p_{26} \oplus p_{27}p_{24} \oplus p_{22} \oplus k_0) \oplus (p_{29} \oplus p_{24} \oplus p_{25}p_{22} \oplus p_{20} \oplus k_4)(p_{24} \oplus p_{19} \oplus p_{20}(p_{17} \oplus p_6 \oplus p_{11}p_9 \oplus p_7p_2 \oplus k_3) \oplus k_{14}) \oplus k_{23}) \oplus (p_5 \oplus (p_{26} \oplus p_{21} \oplus p_{22}p_{19} \oplus (p_{17} \oplus p_6 \oplus p_{11}p_9 \oplus p_7p_2 \oplus k_3) \oplus k_{10}) \oplus (p_{31} \oplus p_{26} \oplus p_{27}p_{24} \oplus p_{22} \oplus k_0)(p_{29} \oplus p_{24} \oplus p_{25}p_{22} \oplus p_{20} \oplus k_4) \oplus (p_{27} \oplus p_{22} \oplus p_{23}p_{20} \oplus (p_{18} \oplus p_7 \oplus p_{12}p_{10} \oplus p_8p_3 \oplus k_1) \oplus k_8)(p_{22} \oplus (p_{17} \oplus p_6 \oplus p_{11}p_9 \oplus p_7p_2 \oplus k_3) \oplus (p_{18} \oplus p_7 \oplus p_{12}p_{10} \oplus p_8p_3 \oplus k_1)(p_{15} \oplus p_4 \oplus p_9p_7 \oplus p_5p_0 \oplus k_7) \oplus k_{18}) \oplus k_{27}) \oplus k_{34} = 0,$$

$$r_{29} = p_{21} \oplus (p_{16} \oplus p_5 \oplus p_{10}p_8 \oplus p_6p_1 \oplus k_5) \oplus (p_{17} \oplus p_6 \oplus p_{11}p_9 \oplus p_7p_2 \oplus k_3)(p_{14} \oplus p_3 \oplus p_8p_6 \oplus p_4(p_{31} \oplus p_{26} \oplus p_{27}p_{24} \oplus p_{22} \oplus k_0) \oplus k_9) \oplus (p_{12} \oplus p_1 \oplus p_6p_4 \oplus p_2(p_{29} \oplus p_{24} \oplus p_{25}p_{22} \oplus p_{20} \oplus k_4) \oplus k_{13}) \oplus k_{20} = 0,$$

$$r_{33} = ((p_{17} \oplus p_6 \oplus p_{11}p_9 \oplus p_7p_2 \oplus k_3) \oplus (p_{12} \oplus p_1 \oplus p_6p_4 \oplus p_2(p_{29} \oplus p_{24} \oplus p_{25}p_{22} \oplus p_{20} \oplus k_4) \oplus k_{13}) \oplus (p_{13} \oplus p_2 \oplus p_7p_5 \oplus p_3(p_{30} \oplus p_{25} \oplus p_{26}p_{23} \oplus p_{21} \oplus k_2) \oplus k_{11})(p_{10} \oplus (p_{31} \oplus p_{26} \oplus p_{27}p_{24} \oplus p_{22} \oplus k_0) \oplus p_4p_2 \oplus p_0(p_{27} \oplus p_{22} \oplus p_{23}p_{20} \oplus (p_{18} \oplus p_7 \oplus p_{12}p_{10} \oplus p_8p_3 \oplus k_1) \oplus k_8) \oplus k_{17}) \oplus k_{28}) = 0.$$

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multi-differential neural distinguishers, we attempt to use $s$ neutral bits to create a plaintext structure, which is expected to pass the initial 26-round differential characteristic together. However, the 26-round differential characteristic is too long to apply for neutral bits. We extend the concept of neutral bits and propose a *weak neutral bit set*.

**Definition 1** *Let $e_0, e_1, \ldots, e_{n-1}$ be the standard basis of $F_2^n$. The input difference of plaintext pair $(P, P')$ is $\Delta_{in}$. If the resulting ciphertext pair is $(C, C')$ and the output difference is $\Delta_{out}$, we claim $(P, P')$ conforms to the differential characteristic $\Delta_{in} \to \Delta_{out}$. Let $I = \{e_i \oplus e_j | i, j \in [0, n-1]\}$, if $(P, P')$ and $(P \oplus I, P' \oplus I)$ conform to the differential characteristic $\Delta_{in} \to \Delta_{out}$ and then $(P \oplus e_i, P' \oplus e_i)$ and $(P \oplus e_j, P' \oplus e_j)$ do not conform to the differential characteristic $\Delta_{in} \to \Delta_{out}$, respectively. We claim $\{i, j\}$ is a weak neutral bit set.*

Plaintext structures are generated by weak neutral bit sets, which pass the initial 26-round differential characteristic together with probability 1. For KATAN32, five bits [30, 29, 26, 14, 12] are only shown in the last condition. Hence, {30, 13}, {29, 13}, {26, 13}, {14, 13}, {12, 13} ($p_{13}$ is changed in the last conditions) are five weak neutral bit sets of KATAN32. Randomly generating a plaintext block $P^{\text{block}}$, we can use five weak neutral bit sets to generate a plaintext structure $P_{\text{structure}}$ consisting of $2^5$ plaintext blocks.

## 5.2 | Training of the conditional neural distinguishers

An $l + r$-round conditional neural distinguisher consists of an $l$-round differential characteristic and an $r$-round multi-differential neural distinguisher. There are two ways to train the conditional neural distinguisher: One way, we train the multi-differential neural distinguisher with $\Delta = 0x00002000$ and add an $l$-round differential characteristic at the key recovery stage; the second way, we use $\Delta P = 0x61082200$ to train the conditional neural distinguisher with conditions directly. Under the same round, the conditional neural distinguisher, which is trained in the second way, has higher accuracy and better wrong key profile. These improvements increase the key recovery attack by 1-2 rounds. The accuracy of the conditional neural distinguishers of KATAN32 are shown in Table 6.

As shown in Table 6, the accuracy of the conditional neural distinguishers decreases with the increasing rounds. We can train an 84-round conditional neural distinguisher at most. After all the preparations, we present a 97-round practical key recovery attack of KATAN32.

**TABLE 6** The accuracy of conditional neural distinguishers of KATAN32

| Round | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 |
|---|---|---|---|---|---|---|---|---|
| Accuracy | 81.6% | 79.4% | 74.5% | 73.3% | 70.2% | 63.7% | 60.6% | 60.2% |

## 5.3 | 97-Round practical key recovery attack of KATAN32

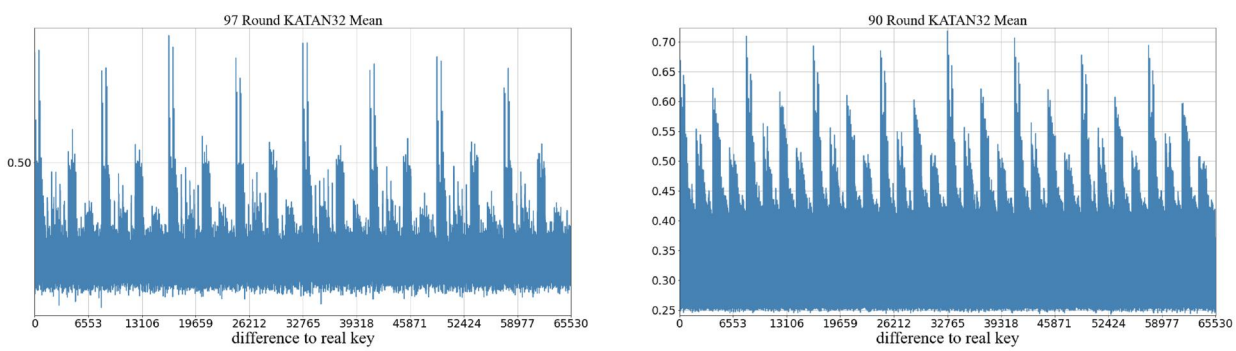
The attack process is as follows:

The plaintext is $P = (p_{31}, p_{30}, \ldots, p_0)$ and the input difference of plaintext is $\Delta P = 0x61082200$. The plaintext pair is $(P, P')$, where $P' = (p'_{31}, p'_{30}, \ldots, p'_0)$. We randomly generate a plaintext block $P^{\text{block}}$ consisting of 32 $(P, P')$. We use five weak neutral bit sets {30, 13}, {29, 13}, {26, 13}, {14, 13}, {12, 13} to generate a plaintext structure $P_{\text{structure}}$ consisting of 32 $P^{\text{block}}$. After the 97-round encryption, we get the resulting ciphertext structure $C_{\text{structure}}$.

Zhang et al. [21] proposed a method to use algebraic expressions to express the dependencies of intermediate states on the outputs. Inspired by this method, the output difference of the 84-round ciphertext is expressed by the 97-round ciphertext and subkeys. The output difference of the 84-round KATAN32 is:

$$\Delta L_{84} = (\Delta l_{84}, \Delta l_{85}, \ldots, \Delta l_{96})$$

$$\Delta R_{84} = (\Delta r_{84}, \Delta r_{85}, \ldots, \Delta r_{102})$$




(a) Wrong key response profile of 97-round KATAN32 with 84-round conditional neural distinguisher

(b) Wrong key response profile of 90-round KATAN32 with 77-round conditional neural distinguisher

**FIGURE 6** Wrong key response profiles of KATAN32

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After all the output differences are expressed by the 97-round ciphertext and subkeys, no subkeys are contained in the expressions of $\Delta r_{97}$–$\Delta r_{102}$. Therefore, 32 expressions only involve 16-bit subkeys $k_{176}$, $k_{178}$, $k_{180}$-$k_{193}$. The wrong key response profiles are shown in Figure 6, which contain the empirical mean and standard deviation. We only show the empirical mean in Figure 6. The wrong key response profile of the 97-round KATAN32 with the 84-round conditional neural distinguisher is shown in Figure 6a.

Using algorithm 1, we partially decrypt the ciphertext structure $C_{\text{structure}}$ with each candidate key and score each candidate key we tried. If the score is higher than a cutoff threshold $c_1$, we use the corresponding key bits $k_{180}$–$k_{193}$ to decrypt seven rounds and get a 90-round ciphertext structure $C'_{structure}$. The process is repeated for another 13 rounds with the 77-round conditional neural distinguisher. The output difference of the 77-round KATAN32 is as follows:

$$\Delta L_{77} = (\Delta l_{77}, \Delta l_{78}, \ldots, \Delta l_{89})$$

$$\Delta R_{77} = (\Delta r_{77}, \Delta r_{78}, \ldots, \Delta r_{95})$$

No subkeys are contained in the expressions of $\Delta r_{90}$–$\Delta r_{95}$. The 32 expressions only involve 16 bit subkeys $k_{162}$, $k_{164}$, $k_{166}$–$k_{179}$. $k_{176}$ and $k_{178}$ have been recovered. Since the accuracy of the 77-round neural distinguisher is higher, the results of $k_{176}$ and $k_{178}$ are determined by the 77-round neural distinguisher. The wrong key response profile for 90-round KATAN32 is shown in Figure 6b. Using one, we partially decrypt the $C'_{structure}$ with each candidate key and score each candidate key we tried. If the score is higher than a cutoff threshold $c_2$, we terminate the search and return the key guess. All subkeys of the last 14 rounds $k_{166}$-$k_{193}$ and part of subkeys of the last 15 and 16 rounds $k_{162}$ and $k_{164}$ can be recovered.

In Ref. [8], Gohr's definition of a successful key recovery attack is that the last-round key is correct and the second-round key is at most two Hamming distance to the real key. Similarly, we count a success if the last 14 key bits are guessed correctly and the second 16 key bits are wrong for no more than two bits. The cutoff threshold parameters of the key recovery attack are $c_1 = 5$, $c_2 = 50$. The key recovery attack is successful in 69 out of 100 trials. The success rate is 69%. In all of these 69 cases, the recovery of the last 14 key bits is successful and the second 16 key bits is wrong for at most two bits.

We analyse the data and time complexity of the attacks. The data complexity is on average $2^{15.5}$ in each attack. For the time complexity, the average time of 100 trials is 2844.4 s. It costs 1.95 s to encrypt $2^{10}$ 97-round KATAN32 for our hardware (Intel Xeon 6226R@2.90Ghz, Nvidia GeForce RTX3090). The time complexity is $2844.4s/\left(\frac{1.95s}{2^{10}}\right) \approx 2^{20.5}$.

To improve the success rate of key recovery attacks and recover the 30 key bits exactly, we run 10 times key recovery attacks with a different $P_{\text{structure}}$ for the same key. For each key bit $k_i$, if the attacks are successful over seven times, we consider that $k_i$ is correctly guessed. The experiment results of KATAN ciphers are shown in Table 7.

As shown in Table 7, all 30 key bits are successfully recovered. To present a full key recovery attack, we do an exhaustive search for the remaining 50 bits. The time complexity is about $2^{50}$. Since the key schedule of KATAN32 uses an 80-bit LFSR, the whole 80-bit master key can be calculated by 80 continuous subkey bits.

## 6 | KEY RECOVERY ATTACK OF KATAN48 AND KATAN64

The key recovery procedures of KATAN48 and KATAN64 are similar to KATAN32. We present the practical key recovery attacks of the 82-round KATAN48 and 70-round KATAN64 in this section.

### 6.1 | 82-Round practical key recovery attack of KATAN48

We train multi-differential neural distinguishers of all input differences with one and two Hamming weights. The $\Delta = 0x040000000000$ is the best input difference. We use the MILP model to search for the smallest number of conditions with the output difference $\Delta = 0x040000000000$. A 25-round differential characteristic with 17 conditions has been searched. After a message modification, we should apply another eight conditions to reduce the computational complexity for filtering the plaintexts. After imposing 25 conditions, we have a 25-round differential characteristic with $\Delta P = 0x610010080402 \xrightarrow{prob.=1} \Delta = 0x040000000000$. We use {46, 44}, {43, 44}, {42, 44}, {24, 44} and {22, 44} to boost the response of the multi-differential neural distinguishers, and add the 25-round differential characteristic before each multi-differential neural distinguisher to get the conditional neural distinguishers. The accuracy of the conditional neural distinguishers of KATAN48 is shown in Table 8.

**TABLE 8** The accuracy of conditional neural distinguishers of KATAN48

| Round | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 |
|---|---|---|---|---|---|---|---|---|
| Accuracy | 95.5% | 90.9% | 84.7% | 79.1% | 71.9% | 66.8% | 63.3% | 58.2% |

**TABLE 7** Key recovery attack results of KATAN32

| key bit | $k_{162}$ | $k_{164}$ | $k_{166}$ | $k_{167}$ | $k_{168}$ | $k_{169}$ | $k_{170}$ | $k_{171}$ | $k_{172}$ | $k_{173}$ | $k_{174}$ | $k_{175}$ | $k_{176}$ | $k_{177}$ | $k_{178}$ | $k_{179}$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Success | 7 | 7 | 8 | 8 | 8 | 8 | 7 | 7 | 8 | 10 | 8 | 10 | 10 | 7 | 9 | 10 |
| **Key bit** | $k_{180}$ | $k_{181}$ | $k_{182}$ | $k_{183}$ | $k_{184}$ | $k_{185}$ | $k_{186}$ | $k_{187}$ | $k_{188}$ | $k_{189}$ | $k_{190}$ | $k_{191}$ | $k_{192}$ | $k_{193}$ | | |
| Success | 10 | 7 | 7 | 10 | 10 | 7 | 10 | 10 | 10 | 10 | 7 | 10 | 10 | 10 | | |

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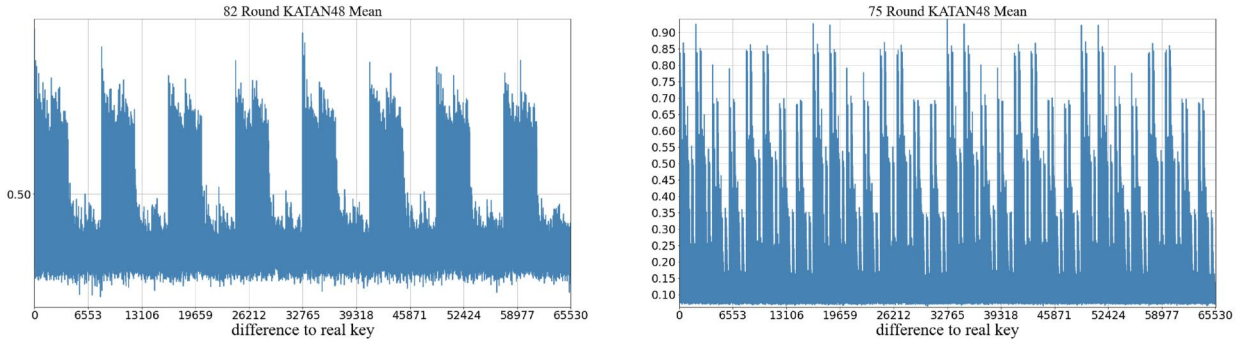


(a) Wrong key response profile of 82-round KATAN48 with 72-round conditional neural distinguisher

(b) Wrong key response profile of 75-round KATAN48 with 65-round conditional neural distinguisher

**FIGURE 7** Wrong key response profiles of KATAN48

**TABLE 9** Key recovery attack results of KATAN48

| **key bit** | $k_{132}$ | $k_{134}$ | $k_{136}$ | $k_{137}$ | $k_{138}$ | $k_{139}$ | $k_{140}$ | $k_{141}$ | $k_{142}$ | $k_{143}$ | $k_{144}$ | $k_{145}$ | $k_{146}$ | $k_{147}$ | $k_{148}$ | $k_{149}$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Success | 7 | 9 | 9 | 10 | 7 | 10 | 9 | 10 | 9 | 10 | 9 | 10 | 10 | 10 | 10 | 10 |
| **Key bit** | $k_{150}$ | $k_{151}$ | $k_{152}$ | $k_{153}$ | $k_{154}$ | $k_{155}$ | $k_{156}$ | $k_{157}$ | $k_{158}$ | $k_{159}$ | $k_{160}$ | $k_{161}$ | $k_{162}$ | $k_{163}$ | | |
| Success | 9 | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 10 | | |

We use 65- and 72-round conditional neural distinguishers to present an 82-round practical key recovery attack of KATAN48. The wrong key response profiles are shown in Figure 7. Using algorithm 1, we can recover all subkeys of the last 14 rounds $k_{136}$-$k_{163}$ and part of subkeys of the last 15 and 16 rounds $k_{132}$ and $k_{134}$. The cutoff threshold parameters of the key recovery attack are $c_1 = 5$, $c_2 = 50$. The key recovery attacks are successful in 84 out of 100 trials. The success rate is 84%. In all of these 84 cases, the recovery of the last 14 key bits is successful and the second 16 key bits are wrong for 2 bits at most. We analyse the data and time complexity of the attacks. The data complexity is $2^{14.2}$ in each trial. The time complexity is $1229.6s/\left(\frac{2.65s}{2^{10}}\right) \approx 2^{18.9}$.

For the same key, we run 10 times key recovery attacks with a different $P_{structure}$. As shown in Table 9, all 30 key bits are successfully recovered. To present a full key recovery attack, we do an exhaustive search for the remaining 50 bits. The time complexity is about $2^{50}$.

## 6.2 | 70-Round practical key recovery attack of KATAN64

The best input difference of KATAN64 is $\Delta = 0x0000000000200000$. We use the MILP model to search for the smallest number of conditions with the output difference $\Delta = 0x0000000000200000$. We get a 25-round differential characteristic with 20 conditions and apply six conditions additionally to reduce the computational complexity. After imposing 26 conditions, we have a 25-round differential characteristic with $\Delta P = 0x2480482010080400 \xrightarrow{prob.=1} \Delta = 0x0000000000200000$. We use {56, 60}, {40, 60}, {38, 60} and {26, 60}, {0, 60} to boost the response of the multi-differential neural distinguishers and train the conditional neural distinguishers. The accuracy of the conditional neural distinguishers of KATAN64 is shown in Table 10.

**TABLE 10** The accuracy of conditional neural distinguishers of KATAN64

| **Round** | **54** | **55** | **56** | **57** | **58** | **59** | **60** | **61** |
|---|---|---|---|---|---|---|---|---|
| Accuracy | 99.7% | 98.2% | 94.5% | 84.5% | 73.1% | 69.4% | 66.3% | 61.3% |

We use 54- and 61-round conditional neural distinguishers to present a 70-round practical key recovery attack of KATAN64. The wrong key response profiles are shown in Figure 8. Using algorithm 1, we can recover all subkeys of the last 15 rounds $k_{110}$-$k_{139}$. The cutoff threshold parameters of the key recovery attack are $c_1 = 0$, $c_2 = 10$. The key recovery attacks are successful in 99 out of 100 trials. The success rate is 99%. The data complexity is $2^{16.7}$ in each attack. The time complexity is $3787.5s/\left(\frac{3.16s}{2^{10}}\right) \approx 2^{20.2}$.

For the same key, we run 10 times key recovery attacks with a different $P_{structure}$. As shown in Table 11, 29 key bits are successfully recovered except $k_{110}$. We then get two candidates for these 30 key bits. To present a full key recovery attack, we do an exhaustive search for the remaining 50 bits. The time complexity is about $2 \times 2^{50} = 2^{51}$.

## 7 | CONCLUSION

In the study, we present a conditional differential analysis on KATAN family based on deep learning. We propose a multi-differential neural distinguisher by modifying the data format and the network structure to improve the accuracy of KATAN neural distinguisher and increase the number of its rounds. The

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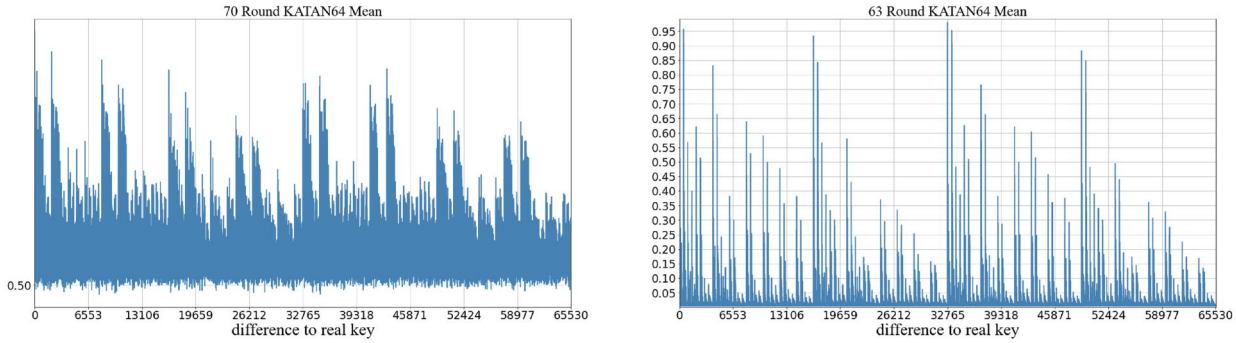


(a) Wrong key response profile of 70-round KATAN64 with 61-round conditional neural distinguisher

(b) Wrong key response profile of 63-round KATAN64 with 54-round conditional neural distinguisher

**FIGURE 8** Wrong key response profiles of KATAN64

**TABLE 11** Key recovery attack results of KATAN64

| **key bit** | $k_{110}$ | $k_{111}$ | $k_{112}$ | $k_{113}$ | $k_{114}$ | $k_{115}$ | $k_{116}$ | $k_{117}$ | $k_{118}$ | $k_{119}$ | $k_{120}$ | $k_{121}$ | $k_{122}$ | $k_{123}$ | $k_{124}$ | $k_{125}$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Success | 5 | 10 | 10 | 10 | 10 | 10 | 9 | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 10 |
| **Key bit** | $k_{126}$ | $k_{127}$ | $k_{128}$ | $k_{129}$ | $k_{130}$ | $k_{131}$ | $k_{132}$ | $k_{133}$ | $k_{134}$ | $k_{135}$ | $k_{136}$ | $k_{137}$ | $k_{138}$ | $k_{139}$ | | |
| Success | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 10 | | |

**TABLE 12** Summary of practical single-key attacks on KATAN ciphers

| Cipher | Technique | Rounds | Recoverable key bits | Time | Data | Reference |
|---|---|---|---|---|---|---|
| KATAN32 | CD | 78 | 4 | $2^{22}$ | $2^{22}$CP | [12] |
| | Cube | 60 | 41 | $2^{39}$ | $2^{30.3}$CP | [13] |
| | Algebraic | 79 | 45 | 14.7 min | 20 CP | [13] |
| | CDDL | 97 | 30 | $2^{20.5}$ | $2^{15.5}$CP | Section 5.3 |
| | Improved CD | 97 | 10 | $2^{30}$ | $2^{19}$ CP | [14] |
| | | 98 | 11 | $2^{31}$ | $2^{19}$ CP | [14] |
| | | 99 | 13 | $2^{33}$ | $2^{19}$ CP | [14] |
| KATAN48 | CD | 70 | 2 | $2^{34}$ | $2^{34}$CP | [12] |
| | Cube | 40 | 31 | $2^{49}$ | $2^{25.9}$CP | [13] |
| | Algebraic | 64 | 40 | 6.4 hours | 5 CP | [13] |
| | CDDL | 82 | 30 | $2^{18.9}$ | $2^{14.2}$CP | Section 6.1 |
| KATAN64 | CD | 68 | 2 | $2^{35}$ | $2^{35}$CP | [12] |
| | Cube | 30 | 25 | $2^{35}$ | $2^{20.7}$ | [13] |
| | Algebraic | 60 | 40 | 3.2 hours | 5 KP | [13] |
| | CDDL | 70 | 30 | $2^{20.2}$ | $2^{16.7}$CP | Section 6.2 |

Abbreviations: CP, chosen plaintext; KP, known plaintext.

conditional neural distinguisher is built consisting of the multi-differential neural distinguisher and conditions, which increases the attack rounds. Moreover, we use message modification technology and a key search strategy to reduce the computational complexity of the key recovery attack. The conditional differential analysis results of KATAN ciphers are improved in the single-key setting. In Table 12, we summarise the results of practical key recovery attacks on KATAN ciphers in the single-key setting.

Compared with the results in Ref. [12–14], our attacks improve the number of recoverable key bits and have advantages in time complexity and data complexity for KATAN32. As shown in Table 12, our work is the best result in the practical key recovery attack for KATAN48 and KATAN64.

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We attempt to present the practical key recovery attacks of KATAN ciphers in the related-key scenario in the future. Applying CDDL to other algorithms is also an interesting topic in the future work.

## AUTHOR CONTRIBUTIONS

**Dongdong Lin**: Conceptualisation; Data curation; Formal analysis; Investigation; Validation; Visualisation; Writing – original draft. **Manman Li**: Writing – review & editing. **Zezhou Hou**: Formal analysis. **Shaozhen Chen**: Writing – review & editing.

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## CONFLICT OF INTEREST

The authors declare that there is no conflict of interest that could be perceived as prejudicing the impartiality of the research reported.

## DATA AVAILABILITY STATEMENT

The data that support the findings of this study are openly available in deepkatan at https://github.com/alin%20dl/deepkatan.

## ORCID

*Dongdong Lin* https://orcid.org/0000-0002-4049-4557
*Manman Li* https://orcid.org/0000-0001-7625-737X
*Zezhou Hou* https://orcid.org/0000-0002-7764-9663
*Shaozhen Chen* https://orcid.org/0000-0003-1015-8875

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